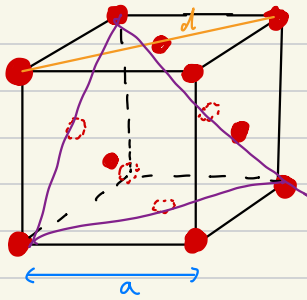


FCC:



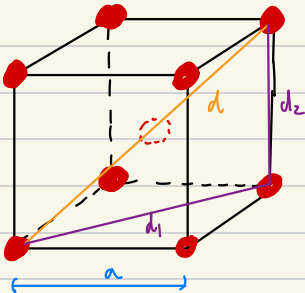
Lattice parameter:

$$\begin{aligned} a^2 + a^2 &= d^2 \\ 2a^2 &= d^2, \quad d = 4R \\ 2a^2 &= 16R^2 \\ a^2 &= 8R^2 \\ a &= 2\sqrt{2}R \end{aligned}$$

APF:

$$\begin{aligned} V_{\text{space}} &= a^3 \\ V_{\text{atom}} &= 4 \times \frac{4}{3}\pi R^3 \\ &= \frac{16}{3}\pi \left(\frac{a}{2\sqrt{2}}\right)^3 \\ \frac{V_a}{V_s} &= \frac{\frac{16}{3}\pi \left(\frac{a^3}{16\sqrt{2}}\right)}{a^3} \\ &= \frac{\pi}{3\sqrt{2}} \\ &\sim 74\% \end{aligned}$$

BCC:



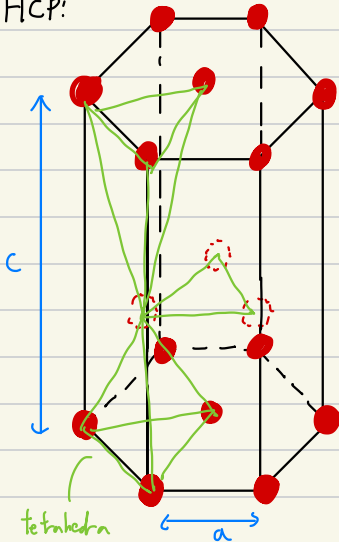
Lattice parameter:

$$\begin{aligned} d^2 &= d_1^2 + d_2^2, \quad d = 4R, \quad d_1^2 = a^2 + a^2, \quad d_2 = a \\ d_1^2 &= 2a^2 \\ (4R)^2 &= 2a^2 + a^2 \\ 16R^2 &= 3a^2 \\ a^2 &= \frac{16}{3}R^2 \\ a &= \frac{4}{\sqrt{3}}R \end{aligned}$$

APF:

$$\begin{aligned} V_{\text{space}} &= a^3 \\ V_{\text{atom}} &= 2 \times \frac{4}{3}\pi R^3 \\ &= 2 \times \frac{4}{3}\pi \left(\frac{a\sqrt{3}}{4}\right)^3 \\ &= \frac{8}{3}\pi \frac{3a^3\sqrt{3}}{64} \\ &= \frac{\pi\sqrt{3}a^3}{8} \\ \frac{V_a}{V_s} &= \frac{\frac{\pi\sqrt{3}a^3}{8}}{a^3} = \frac{\pi\sqrt{3}}{8} \\ &\sim 68\% \end{aligned}$$

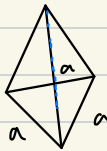
HCP:



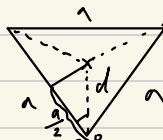
Lattice parameter:

$$a: 2r$$

c:

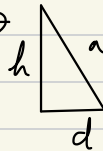


Consider this Δ :



$$\begin{aligned} \cos 30^\circ &= \frac{a}{2h} \\ \frac{\sqrt{3}}{2} &= \frac{a}{2h} \\ d &= \frac{a}{\sqrt{3}} \end{aligned}$$

Now consider this Δ :

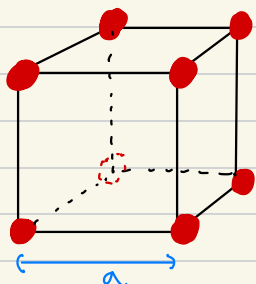


$$\begin{aligned} h^2 + d^2 &= a^2 \\ h^2 &= a^2 - d^2 \\ h^2 &= a^2 - \frac{a^2}{3} \\ h^2 &= \frac{2a^2}{3} \\ h &= \sqrt{\frac{2}{3}}a \\ c &= 2h \\ &= 2\sqrt{\frac{2}{3}}a \end{aligned}$$

APF:

$$\begin{aligned} V_{\text{cell}} &= a^2 c \\ \cos 30^\circ &= \frac{h}{a} \\ h &= \frac{a\sqrt{3}}{2} \\ A_{\Delta} &= \frac{ah}{2} = \frac{a^2\sqrt{3}}{4} \\ A_{\square} &= 6 \times \frac{a^2\sqrt{3}}{4} = \frac{3\sqrt{3}}{2}a^2 \\ V_c &= \frac{3\sqrt{3}}{2}a^2(c) \\ &= \frac{3\sqrt{3}}{2}a^2 \cdot 2\sqrt{\frac{2}{3}}a \\ &= 3\sqrt{2}a^3 \\ V_{\text{atom}} &= 6 \times \frac{4}{3}\pi R^3 \\ &= 6 \times \frac{4}{3}\pi \left(\frac{a}{2}\right)^3 \\ &= a^3\pi \\ \frac{V_a}{V_s} &= \frac{a^3\pi}{3\sqrt{2}a^3} = \frac{\pi}{3\sqrt{2}} \sim 74\% \end{aligned}$$

Simple cubic:



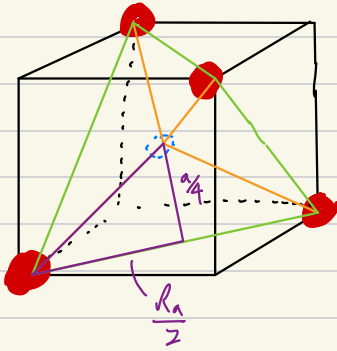
Lattice parameter:

$$a = 2R$$

APF

$$\begin{aligned} V_{\text{space}} &= a^3 \\ V_{\text{atom}} &= \frac{4}{3}\pi R^3 \\ &= \frac{4}{3}\pi \left(\frac{a}{2}\right)^3 \\ &= \frac{\pi a^3}{6} \\ \frac{V_a}{V_s} &= \frac{\pi}{6} \sim 52\% \end{aligned}$$

FCC tetrahedral



$$\left(\frac{a}{4}\right)^2 + (r_A)^2 = (r_A + r_C)^2$$

$$\left(\frac{2\sqrt{2}}{4} r_A\right)^2 + (r_A)^2 = (r_A + r_C)^2$$

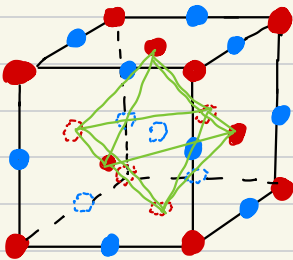
$$r_A^2 \left(\frac{8}{16} + 1 \right) = (r_A + r_C)^2$$

$$\frac{3}{2} r_A^2 = (r_A + r_C)^2$$

$$\frac{\sqrt{3}}{2} r_A = r_A + r_C$$

$$r_C = \left(\frac{\sqrt{3}}{2} - 1\right) r_A \sim 0.225 r_A$$

FCC octahedral



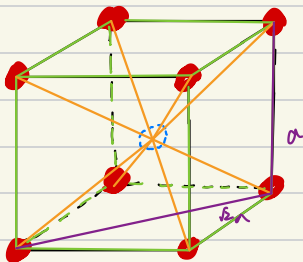
$$2(r_A + r_C) = a$$

$$2(r_A + r_C) = 2\sqrt{2} r_A$$

$$r_A + r_C = \sqrt{2} r_A$$

$$r_C = (\sqrt{2} - 1) r_A \sim 0.414 r_A$$

Simple Cubic, single cubic.



$$2(r_A + r_C) = \sqrt{(\sqrt{2}a)^2 + a^2}$$

$$2(r_A + r_C) = \sqrt{3} a$$

$$, a = 2r_A \text{ (simple cubic)}$$

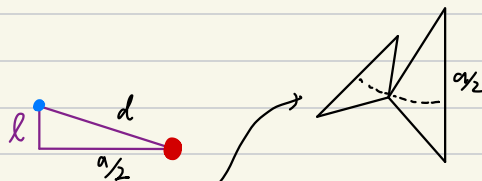
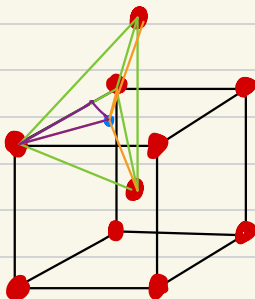
$$2(r_A + r_C) = \sqrt{3} (2r_A)$$

$$r_A + r_C = \sqrt{3} r_A$$

$$r_C = (\sqrt{3} - 1) r_A$$

$$\sim 0.732 r_A$$

BCC tetrahedral



$$l = \frac{a}{4}$$

$$d^2 = \left(\frac{a}{4}\right)^2 + \left(\frac{a}{4}\right)^2$$

$$d^2 = \frac{5}{16} a^2$$

$$d^2 = \frac{5}{16} \left(\frac{4}{\sqrt{3}}\right)^2 r_A^2$$

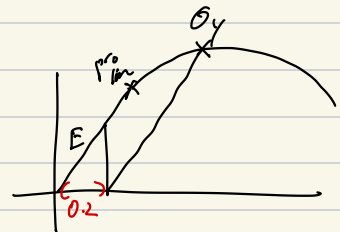
$$d^2 = \frac{5}{16} \left(\frac{16}{3}\right) r_A^2$$

$$d^2 = \frac{5}{3} r_A^2$$

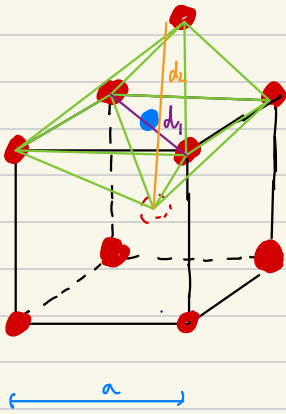
$$d = \sqrt{\frac{5}{3}} r_A$$

$$r_A + r_C = \sqrt{\frac{5}{3}} r_A$$

$$r_C = \left(\sqrt{\frac{5}{3}} - 1\right) r_A$$



BCC octahedral



$$d_1 = 2(R_A + R_C) \text{ and } d_1^2 = 2a^2$$

$$d_1 = \sqrt{2}a$$

$$\sqrt{2}a = 2(R_A + R_C)$$

$$\sqrt{2} \left(\frac{4}{\sqrt{3}} R_A \right) = 2(R_A + R_C)$$

$$\frac{4}{\sqrt{6}} R_A = R_A + R_C$$

$$R_{C1} = R_A \left(\frac{4}{\sqrt{6}} - 1 \right)$$

$$d_2 = 2(R_A + R_C) \text{ and } d_2 = a$$

$$\frac{4}{\sqrt{3}} R_A = 2(R_A + R_C)$$

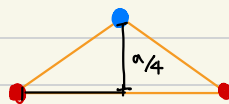
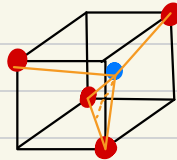
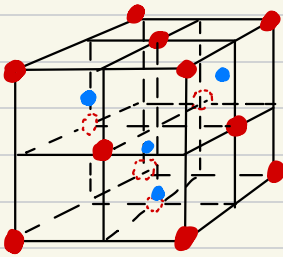
$$R_C = \frac{2}{\sqrt{3}} R_A - R_A$$

$$R_{C2} = \left(\frac{2}{\sqrt{3}} - 1 \right) R_A$$

$$R_{C2} < R_{C1}$$

$$\therefore R_C = \left(\frac{2}{\sqrt{3}} - 1 \right) R_A$$

Diamond Cubic:



$$h^2 = \left(\frac{a}{2} \right)^2 + \left(\frac{a}{2} \right)^2$$

$$h^2 = \frac{a^2}{2}$$

$$h = \frac{a}{\sqrt{2}}$$

$$\frac{h}{2} = \frac{a}{2\sqrt{2}}$$

$$\left(\frac{a}{2\sqrt{2}} \right)^2 + \left(\frac{a}{4} \right)^2 = (2R)^2$$

$$\frac{a^2}{8} + \frac{a^2}{16} = 4R^2$$

$$\frac{3a^2}{16} = 4R^2$$

$$a^2 = \frac{64}{3} R^2$$

$$a = \frac{8}{\sqrt{3}} R$$

APF:

$$\frac{8 \left(\frac{4}{3} \pi R^3 \right)}{a^3}$$

$$= \frac{8 \cdot \frac{4}{3} \pi R^3}{\left(\frac{8}{\sqrt{3}} \right)^3}$$

$$= \frac{32\pi R^3}{\frac{512}{\sqrt{3}}}$$

$$= \frac{32\pi \sqrt{3}}{512}$$

$$= \frac{\sqrt{3}\pi}{16} \sim 34\%$$

$$CN = 4$$

$$\text{Atm pr cell: } 8$$

$$\frac{16}{192}$$